

Exponential Dispersion Family

Definition 1.1. An **exponential dispersion density** takes the form:

$$f(y_i|\theta_i, \phi) = \exp \left\{ \frac{y_i\theta_i - b(\theta_i)}{\phi} + c(y_i, \phi) \right\},$$

where θ_i is the *canonical parameter*, ϕ is the *dispersion parameter*, $b(\cdot)$ is the *cumulant function*, and $c(y_i, \phi)$ is the log partition function. ■

Proposition 1.1 (Exponential Dispersion Properties).

- The log likelihood contribution of y_i is:

$$\ell(\theta_i, \phi) = \frac{y_i\theta_i - b(\theta_i)}{\phi} + c(y_i, \phi).$$

- The score contribution of y_i :

$$u_i(\theta_i, \phi) = \frac{\partial \ell_i}{\partial \theta_i} = \frac{y_i - \dot{b}(\theta_i)}{\phi}.$$

- The information contribution of y_i :

$$\mathcal{I}_{\theta_i\theta_i} = -\mathbb{E} \left(\frac{\partial^2 \ell_i}{\partial \theta_i^2} \right) = \frac{\ddot{b}(\theta_i)}{\phi}.$$

- The mean $\mathbb{E}(y_i)$ of an exponential dispersion model is the first derivative of the cumulant function:

$$\mu_i = \dot{b}(\theta_i).$$

- The variance of an exponential dispersion model is a function of the mean:

$$\mathbb{V}(y_i) = \phi \ddot{b}(\theta_i) = \phi \ddot{b} \circ \dot{b}^{-1}(\mu_i) \equiv \phi \nu(\mu_i).$$

Here $\nu(\mu_i) = \ddot{b} \circ \dot{b}^{-1}(\mu_i)$ is the *variance function*. ◆

Generalized Linear Models

Definition 2.1. In a **generalized linear model** (GLM), a regression function is specified for the conditional mean:

$$\mathbb{E}(y_i | \mathbf{x}_i) \equiv \mu_i = h(\eta_i).$$

Here μ_i is the conditional mean, $\eta_i = \mathbf{x}_i' \boldsymbol{\beta}$ is the *linear predictor*, and h is the *activation function*. The *link function* g is the inverse of the activation function. The link function applied to the conditional mean returns the linear predictor:

$$g(\mu_i) = \eta_i = \mathbf{x}_i' \boldsymbol{\beta}.$$

The conditional mean model implies a model for the canonical parameter:

$$\theta_i = \dot{b}^{-1}(\mu_i) = (\dot{b} \circ h)(\eta_i).$$

■

2.1 Miscellaneous Relations

Proposition 2.1. The derivative of the activation function is the reciprocal of the derivative of the link function:

$$\dot{h}(\eta_i) = \frac{1}{\dot{g}(\mu_i)}.$$

◆

Proof.

$$1 = \frac{\partial}{\partial \eta_i} \eta_i = \frac{\partial}{\partial \eta_i} g \circ h(\eta_i) = \dot{g}\{h(\eta_i)\} \dot{h}(\eta_i) \implies \dot{h}(\eta_i) = \frac{1}{\dot{g}\{h(\eta_i)\}}.$$

■

Definition 2.2. The canonical parameter is related to the linear predictor via:

$$\theta_i = \dot{b}^{-1} \circ h(\eta_i).$$

If the link function g is selected to coincide with the inverse of the derivative of the cumulant function $g = \dot{b}^{-1}$, such that $h = \dot{b}$, then:

$$\theta_i = \dot{b}^{-1} \circ \dot{b}(\eta_i) = \eta_i.$$

This choice of g is referred to as the **canonical link**. Under the canonical link, the canonical parameter is exactly the linear predictor. ■

Proposition 2.2. Under the canonical link $g = \dot{b}^{-1}$:

$$\nu(\mu_i)\dot{g}(\mu_i) = 1.$$

◆

Proof. Recall that $\dot{b}(\theta_i) = \mu_i$ and $\ddot{b}(\theta_i) = \nu(\mu_i)$. Under the canonical link $h = \dot{b}$ and $\theta_i = \eta_i$, therefore:

$$\nu(\mu_i) = \ddot{b}(\theta_i) = \dot{h}(\eta_i) = \frac{1}{\dot{g}(\mu_i)}.$$

■

Proposition 2.3. For any link function:

$$\frac{\partial \theta_i}{\partial \mu_i} = \frac{1}{\ddot{b}(\theta_i)} = \frac{1}{\nu(\mu_i)}.$$

In the case of the canonical link: $\partial_\mu \theta_i = \dot{g}(\mu_i)$.

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Proof. By implicit differentiation of $\dot{b}(\theta_i) = \mu_i$:

$$\ddot{b}(\theta_i) \frac{\partial \theta_i}{\partial \mu_i} = \frac{\partial \mu_i}{\partial \mu_i} = 1.$$

■

2.2 Properties of GLMs

Proposition 2.4 (GLM Properties). Suppose $\eta_i = \mathbf{x}_i' \boldsymbol{\beta}$.

- The score for $\boldsymbol{\beta}$ is:

$$\mathcal{U}_{\boldsymbol{\beta}} = \sum_{i=1}^n \frac{y_i - \dot{b}(\theta_i)}{\phi \nu(\mu_i)} \frac{\mathbf{x}_i}{\dot{g}(\mu_i)}.$$

- The score for ϕ is:

$$\mathcal{U}_{\phi} = \sum_{i=1}^n \left\{ \frac{\partial c(y_i, \phi)}{\partial \phi} - \frac{y_i \theta_i - b(\theta_i)}{\phi^2} \right\}.$$

- The information for $\boldsymbol{\beta}$ is:

$$\mathcal{I}_{\boldsymbol{\beta}\boldsymbol{\beta}'} = \sum_{i=1}^n \frac{\mathbf{x}_i \mathbf{x}_i'}{\phi \nu(\mu_i) \dot{g}^2(\mu_i)}.$$

- The information for ϕ is:

$$\mathcal{I}_{\phi\phi} = -\sum_{i=1}^n \mathbb{E} \left\{ \frac{\partial^2 c(y_i, \phi)}{\partial \phi^2} \right\} - 2 \sum_{i=1}^n \frac{\mu_i \theta_i - b(\theta_i)}{\phi^3}.$$

- The cross information between β and ϕ is:

$$\mathcal{I}_{\beta\phi} = \mathbf{0}.$$

◆

Proof. The model log likelihood is:

$$\ell(\beta, \phi) = \sum_{i=1}^n \ell_i(\beta, \phi) = \sum_{i=1}^n \frac{y_i \theta_i - b(\theta_i)}{\phi} + c(y_i, \phi).$$

The score for β is:

$$\mathcal{U}_\beta = \frac{\partial \ell}{\partial \beta} = \sum_{i=1}^n \frac{\partial \ell_i}{\partial \theta_i} \frac{\partial \theta_i}{\partial \mu_i} \frac{\partial \mu_i}{\partial \eta_i} \frac{\partial \eta_i}{\partial \beta} = \sum_{i=1}^n \frac{y_i - \dot{b}(\theta_i)}{\phi} \cdot \frac{1}{\ddot{b}(\theta_i)} \cdot \dot{h}(\eta_i) \cdot \mathbf{x}_i.$$

Since $\ddot{b}(\theta_i) = \nu(\mu_i)$:

$$\mathcal{U}_\beta = \sum_{i=1}^n u_i(\beta, \phi) = \sum_{i=1}^n \frac{y_i - \dot{b}(\theta_i)}{\phi \nu(\mu_i)} \frac{\mathbf{x}_i}{\dot{g}(\mu_i)}.$$

The score for ϕ is:

$$\mathcal{U}_\phi = \frac{\partial \ell}{\partial \phi} = \sum_{i=1}^n \frac{\partial \ell_i}{\partial \phi} = \sum_{i=1}^n \left\{ \frac{\partial c(y_i, \phi)}{\partial \phi} - \frac{y_i \theta_i - b(\theta_i)}{\phi^2} \right\}.$$

Towards finding the information for β , observe that $\mathcal{U}_\beta$ is expressed as a function of θ_i and μ_i , each of which is a function of β . Now, the Hessian for β is:

$$\mathcal{H}_{\beta\beta'} = \frac{\partial^2 \ell}{\partial \beta \partial \beta'} = \sum_{i=1}^n \left(\frac{\partial u_i}{\partial \theta_i} \frac{\partial \theta_i}{\partial \mu_i} \frac{\partial \mu_i}{\partial \eta_i} \frac{\partial \eta_i}{\partial \beta'} + \frac{\partial u_i}{\partial \mu_i} \frac{\partial \mu_i}{\partial \eta_i} \frac{\partial \eta_i}{\partial \beta'} \right).$$

Evaluating the first derivative within the sum:

$$\frac{\partial u_i}{\partial \theta_i} \frac{\partial \theta_i}{\partial \mu_i} \frac{\partial \mu_i}{\partial \eta_i} \frac{\partial \eta_i}{\partial \beta} = -\frac{\ddot{b}(\theta_i)}{\phi \nu(\mu_i)} \frac{\mathbf{x}_i}{\dot{g}(\mu_i)} \cdot \frac{1}{\ddot{b}(\theta_i)} \cdot \frac{1}{\dot{g}(\mu_i)} \mathbf{x}_i' = \frac{-\mathbf{x}_i \mathbf{x}_i'}{\phi \nu(\mu_i) \dot{g}^2(\mu_i)}.$$

Observe that the second derivative within the sum is of the form:

$$\{y_i - \dot{b}(\theta_i)\} \frac{\mathbf{x}_i}{w\phi} \frac{\partial}{\partial \mu_i} \frac{1}{\nu(\mu_i) \dot{g}(\mu_i)} \cdot \frac{\partial \mu_i}{\partial \eta_i} \frac{\partial \eta_i}{\partial \beta'}$$

Upon taking the expectation, this term vanishes due to the leading factor of $\{y_i - \dot{b}(\theta_i)\}$. Therefore, the Fisher information for $\boldsymbol{\beta}$ is:

$$\mathcal{I}_{\boldsymbol{\beta}\boldsymbol{\beta}'} = -\mathbb{E} \left(\frac{\partial^2 \ell}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}'} \right) = \sum_{i=1}^n \frac{\mathbf{x}_i \mathbf{x}_i'}{\phi \nu(\mu_i) \dot{g}^2(\mu_i)}.$$

The Hessian for ϕ is:

$$\mathcal{H}_{\phi\phi} = \frac{\partial^2 \ell}{\partial \phi^2} = \sum_{i=1}^n \left\{ \frac{\partial^2 c(y_i, \phi)}{\partial \phi^2} + 2 \frac{y_i \theta_i - b(\theta_i)}{\phi^3} \right\}.$$

The Fisher information for ϕ is:

$$\mathcal{I}_{\phi\phi} = -\mathbb{E} \left(\frac{\partial^2 \ell}{\partial \phi^2} \right) = -\frac{\partial^2 \ell}{\partial \phi^2} = -\sum_{i=1}^n \mathbb{E} \left\{ \frac{\partial^2 c(y_i, \phi)}{\partial \phi^2} \right\} - 2 \sum_{i=1}^n \frac{\mu_i \theta_i - b(\theta_i)}{\phi^3}.$$

The Hessian between $\boldsymbol{\beta}$ and ϕ is:

$$\mathcal{H}_{\boldsymbol{\beta}\phi} = \frac{\partial^2 \ell}{\partial \boldsymbol{\beta} \partial \phi} = \sum_{i=1}^n \frac{\{y_i - \dot{b}(\theta_i)\}}{\phi^2 \nu(\mu_i)} \frac{\mathbf{x}_i}{\dot{g}(\mu_i)}.$$

Due to the factor of $\{y_i - \dot{b}(\theta_i)\}$, the Fisher information between $\boldsymbol{\beta}$ and ϕ vanishes:

$$\mathcal{I}_{\boldsymbol{\beta}\phi} = -\mathbb{E} \left(\frac{\partial^2 \ell}{\partial \boldsymbol{\beta} \partial \phi} \right) = \mathbf{0}.$$

■

Remark 2.1. Since $\hat{\boldsymbol{\beta}}$ and $\hat{\phi}$ are asymptotically independent, a consistent estimate of $\boldsymbol{\beta}$ is obtained by solving the score equations $\mathcal{U}_{\boldsymbol{\beta}}$ for $\boldsymbol{\beta}$ with ϕ replaced by any consistent estimator $\hat{\phi}$. ♦

Proposition 2.5. Define the following $n \times n$ characteristic matrices:

$$\begin{aligned} \boldsymbol{\Delta} &= \text{diag}\{\dot{g}(\mu_i)\}, \\ \mathbf{W} &= \text{diag}\left\{ \frac{1}{\phi \nu(\mu_i) \dot{g}^2(\mu_i)} \right\}, \\ \boldsymbol{\Sigma} &= \text{diag}\{\mathbb{V}(y_i)\} = \text{diag}\{\phi \nu(\mu_i)\}. \end{aligned}$$

These matrices are related via:

$$\mathbf{W}^{-1} = \boldsymbol{\Delta} \boldsymbol{\Sigma} \boldsymbol{\Delta}.$$

Using these forms, the score for $\boldsymbol{\beta}$ is expressible as:

$$\mathcal{U}_{\boldsymbol{\beta}} = \mathbf{X}' \mathbf{W} \boldsymbol{\Delta} (\mathbf{y} - \boldsymbol{\mu}).$$

The information for $\boldsymbol{\beta}$ is expressible as:

$$\mathcal{I}_{\boldsymbol{\beta}\boldsymbol{\beta}'} = \mathbf{X}' \mathbf{W} \mathbf{X}.$$

♦

Corollary 2.1. Under the canonical link $\nu(\mu_i)\dot{g}(\mu_i) = 1$, hence:

$$\mathbf{W} = \text{diag} \left\{ \frac{1}{\phi \dot{g}(\mu_i)} \right\}, \quad \mathbf{W} \Delta = \phi^{-1} \mathbf{I}, \quad \mathbf{W} = \Sigma^{-1}.$$



Result 2.1 (Iteratively Reweighted Least Squares). Suppose $\hat{\beta}^{(r)}$ is the current estimate of β , and define the *working response vector* as:

$$\tilde{\mathbf{y}}^{(r)} = \mathbf{X} \hat{\beta}^{(r)} + \Delta^{(r)} (\mathbf{y} - \boldsymbol{\mu}^{(r)}).$$

The Newton-Raphson update for β is the weighted least squares (WLS) estimator for regression of $\tilde{\mathbf{y}}^{(r)}$ on $\mathbf{X}$ using weights $\mathbf{W}^{(r)}$. That is:

$$\hat{\beta}^{(r+1)} = (\mathbf{X}' \mathbf{W}^{(r)} \mathbf{X})^{-1} \mathbf{X}' \mathbf{W}^{(r)} \tilde{\mathbf{y}}^{(r)}. \quad (2.1)$$



Proof. The Newton-Raphson update for β is:

$$\hat{\beta}^{(r+1)} = \hat{\beta}^{(r)} + \mathcal{I}_{\beta\beta'}^{-1}(\hat{\beta}^{(r)}) \mathcal{U}_{\beta}(\hat{\beta}^{(r)}).$$

Writing out the score and information:

$$\begin{aligned} \hat{\beta}^{(r+1)} &= \hat{\beta}^{(r)} + (\mathbf{X}' \mathbf{W}^{(r)} \mathbf{X})^{-1} \mathbf{X}' \mathbf{W}^{(r)} \Delta^{(r)} (\mathbf{y} - \boldsymbol{\mu}^{(r)}) \\ &= (\mathbf{X}' \mathbf{W}^{(r)} \mathbf{X})^{-1} \left\{ (\mathbf{X}' \mathbf{W}^{(r)} \mathbf{X}) \hat{\beta}^{(r)} + \mathbf{X}' \mathbf{W}^{(r)} \Delta^{(r)} (\mathbf{y} - \boldsymbol{\mu}^{(r)}) \right\} \\ &= (\mathbf{X}' \mathbf{W}^{(r)} \mathbf{X})^{-1} \mathbf{X}' \mathbf{W}^{(r)} \left\{ \mathbf{X} \hat{\beta}^{(r)} + \Delta^{(r)} (\mathbf{y} - \boldsymbol{\mu}^{(r)}) \right\}. \end{aligned}$$



2.3 Deviance

Definition 2.3. Let $\ell(\boldsymbol{\mu}, \phi; \mathbf{y})$ denote the log likelihood as a function of the mean vector $\boldsymbol{\mu} = (\mu_1, \dots, \mu_n)$ and the dispersion parameter ϕ . If $\hat{\beta}$ is the MLE of β and $\hat{\boldsymbol{\mu}} = h(\mathbf{X} \hat{\beta})$, then $\ell(\hat{\boldsymbol{\mu}}, \phi; \mathbf{y})$ is the realized log likelihood. The maximum attainable log likelihood is $\ell(\mathbf{y}, \phi; \mathbf{y})$. Let $\hat{\theta}_i$ denote the canonical parameter for the i th observation under the MLE, and let $\tilde{\theta}_i$ denote the canonical parameter for the model that maximizes the log likelihood. The **scaled deviance** is:

$$D = -2 \{ \ell(\hat{\boldsymbol{\mu}}, \phi; \mathbf{y}) - \ell(\mathbf{y}, \phi; \mathbf{y}) \} = \frac{2}{\phi} \sum_{i=1}^n \left[y_i (\tilde{\theta}_i - \hat{\theta}_i) - \{ b(\tilde{\theta}_i) - b(\hat{\theta}_i) \} \right].$$



Result 2.2. The Pearson χ^2 statistic for GLMs is:

$$T = \sum_{i=1}^n \left\{ \frac{y_i - \mu_i}{\sqrt{\mathbb{V}(y_i)}} \right\}^2 = \sum_{i=1}^n \frac{(y_i - \mu_i)^2}{\phi \nu(\mu_i)} \xrightarrow{\mathcal{L}} \chi_{n-p}^2,$$

where $p = \dim(\boldsymbol{\beta})$. Setting $T \stackrel{\text{Set}}{=} \mathbb{E}\{\chi_{n-p}^2\} = (n-p)$ and solving for ϕ gives a method of moments estimator for ϕ :

$$\hat{\phi} = \frac{1}{n-p} \sum_{i=1}^n \frac{(y_i - \hat{\mu}_i)^2}{\nu(\hat{\mu}_i)}. \quad (2.2)$$



2.4 Quasi Likelihood

Definition 2.4. The **log quasi likelihood** of an observation y_i with mean μ_i :

$$q_i = q(\mu_i) = \int_{y_i}^{\mu_i} \frac{y_i - u}{\phi \nu(\mu_i)} du.$$



Discussion 2.1. The use of quasi likelihood allows for specification of GLMs with non-standard mean-variance relationships. The derivative of q_i is the quasi score:

$$u_i = \frac{\partial q_i}{\partial \mu_i} = \frac{y_i - \mu_i}{\phi \nu(\mu_i)}.$$

Observe that the quasi score has expectation zero and variance:

$$\mathbb{V}(u_i) = \mathbb{V}\left(\frac{y_i - \mu_i}{\phi \nu(\mu_i)}\right) = \frac{1}{\phi \nu(\mu_i)},$$

which coincides with the negative expected Hessian:

$$-\mathbb{E}\left(\frac{\partial^2 q_i}{\partial \mu_i^2}\right) = \frac{1}{\phi \nu(\mu_i)}.$$

The regression parameters are estimated by solving the estimating equations:

$$\mathcal{U}_{\boldsymbol{\beta}} = \frac{\partial \boldsymbol{\mu}'}{\partial \boldsymbol{\beta}} \text{diag}\{\mathbb{V}(y_i)\} \frac{\mathbf{y} - \boldsymbol{\mu}}{\phi} \stackrel{\text{Set}}{=} \mathbf{0},$$

where the dispersion parameter ϕ is estimated as in (2.2).



Inference

Example 3.1. Consider a GLM with linear predictor:

$$\eta_i = \mathbf{x}_i' \boldsymbol{\alpha} + \mathbf{z}_i' \boldsymbol{\beta}.$$

Suppose the hypothesis of interest is $H_0 : \boldsymbol{\beta} = \mathbf{0}$. Let $\hat{\boldsymbol{\alpha}}$ and $\hat{\boldsymbol{\beta}}$ denote solutions to the score equations under the full model:

$$\mathcal{U}_{\alpha}(\boldsymbol{\alpha}, \boldsymbol{\beta}) = \mathbf{X}' \mathbf{W} \boldsymbol{\Delta}(\mathbf{y} - \boldsymbol{\mu}) \stackrel{\text{Set}}{=} \mathbf{0},$$

$$\mathcal{U}_{\beta}(\boldsymbol{\alpha}, \boldsymbol{\beta}) = \mathbf{Z}' \mathbf{W} \boldsymbol{\Delta}(\mathbf{y} - \boldsymbol{\mu}) \stackrel{\text{Set}}{=} \mathbf{0}.$$

Let $\tilde{\boldsymbol{\alpha}}$ denote a solution to the score equation under the null model:

$$\mathcal{U}_{\alpha}(\boldsymbol{\alpha}, \boldsymbol{\beta} = \mathbf{0}) = \mathbf{X}' \mathbf{W} \boldsymbol{\Delta}(\mathbf{y} - \boldsymbol{\mu}) \stackrel{\text{Set}}{=} \mathbf{0},$$

The efficient information for $\boldsymbol{\beta}$ is:

$$\mathcal{I}_{\beta\beta'|\alpha} = \mathbf{Z}' \mathbf{W} \mathbf{Z} - \mathbf{Z}' \mathbf{W} \mathbf{X} (\mathbf{X}' \mathbf{W} \mathbf{X})^{-1} \mathbf{X}' \mathbf{W} \mathbf{Z}. \quad (3.1)$$

The Wald statistic for evaluating $H_0 : \boldsymbol{\beta} = \mathbf{0}$ is:

$$T_W = \hat{\boldsymbol{\beta}}' \mathcal{I}_{\beta\beta'|\alpha} \hat{\boldsymbol{\beta}} \quad (3.2)$$

Let $\tilde{\mathcal{U}}_{\beta}$ denote the score

$$\mathcal{U}_{\beta} = \mathbf{Z}' \mathbf{W} \boldsymbol{\Delta}(\mathbf{y} - \boldsymbol{\mu})$$

for $\boldsymbol{\beta}$ evaluated at $\boldsymbol{\alpha} = \tilde{\boldsymbol{\alpha}}$ and $\boldsymbol{\beta} = \mathbf{0}$. The score statistic for evaluating $H_0 : \boldsymbol{\beta} = \mathbf{0}$ is:

$$T_S = \tilde{\mathcal{U}}_{\beta}' \mathcal{I}_{\beta\beta'|\alpha}^{-1} \tilde{\mathcal{U}}_{\beta}. \quad (3.3)$$

Under H_0 , T_W and T_S are asymptotically $\chi^2_{\dim(\boldsymbol{\beta})}$. ♠

Example 3.2. Consider again a GLM with linear predictor:

$$\eta_i = \mathbf{x}_i' \boldsymbol{\alpha} + \mathbf{z}_i' \boldsymbol{\beta}.$$

A score statistic for testing $H_0 : \boldsymbol{\beta} = \mathbf{0}$ may be constructed using the working vector:

$$\tilde{\mathbf{y}} = \mathbf{X} \boldsymbol{\alpha} + \boldsymbol{\Delta}(\mathbf{y} - \boldsymbol{\mu}).$$

The expectation and variance of $\tilde{\mathbf{y}}$ under H_0 are:

$$\mathbb{E}_0(\tilde{\mathbf{y}}) = \mathbf{X} \boldsymbol{\alpha}, \quad \mathbb{V}_0(\tilde{\mathbf{y}}) = \boldsymbol{\Delta} \boldsymbol{\Sigma} \boldsymbol{\Delta} = \mathbf{W}^{-1}.$$

The GLS estimator of α is:

$$\tilde{\alpha} = (\mathbf{X}'\mathbf{W}\mathbf{X})^{-1}\mathbf{X}'\mathbf{W}\tilde{\mathbf{y}}.$$

The score for β is expressible as:

$$\mathcal{U}_\beta = \mathbf{Z}'\mathbf{W}\Delta(\mathbf{y} - \mu) = \mathbf{Z}'\mathbf{W}(\tilde{\mathbf{y}} - \mathbf{X}\alpha).$$

Evaluating the score at $\alpha = \tilde{\alpha}$ gives:

$$\tilde{\mathcal{U}}_\beta = \mathbf{Z}'\mathbf{Q}\tilde{\mathbf{y}},$$

where $\mathbf{Q} = \mathbf{I} - \mathbf{W}\mathbf{X}(\mathbf{X}'\mathbf{W}\mathbf{X})^{-1}\mathbf{X}'\mathbf{W}$. The variance of $\tilde{\mathcal{U}}_\beta$ is:

$$\mathbb{V}(\tilde{\mathcal{U}}_\beta) = \mathbf{Z}'\mathbf{Q}\mathbf{W}^{-1}\mathbf{Q}\mathbf{Z} = \mathbf{Z}'\mathbf{Q}\mathbf{Z},$$

which is exactly the efficient information (3.1). The working vector score statistic:

$$\begin{aligned} T_S &= \tilde{\mathcal{U}}'_\beta (\mathbf{Z}'\mathbf{Q}\mathbf{Z})^{-1} \tilde{\mathcal{U}}_\beta = \tilde{\mathbf{y}}'\mathbf{Q}\mathbf{Z}(\mathbf{Z}'\mathbf{Q}\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{Q}\tilde{\mathbf{y}} \\ &= (\tilde{\mathbf{y}} - \mathbf{X}\tilde{\alpha})'\mathbf{W}\mathbf{Z}(\mathcal{I}_{\beta|\alpha})^{-1}\mathbf{Z}\mathbf{W}(\tilde{\mathbf{y}} - \mathbf{X}\tilde{\alpha}) \end{aligned}$$

coincides with the standard score statistic in (3.3). 

Examples

4.1 Normal Distribution

Example 4.1 (Canonical Link). The normal distribution takes the form:

$$f(y_i|\mu_i, \sigma^2) = (2\pi\sigma^2)^{-1/2} \exp\left\{-\frac{(y_i - \mu_i)^2}{2\sigma^2}\right\} = \exp\left\{\frac{y_i\mu_i - \mu_i^2/2}{\sigma^2} - \frac{1}{2}\ln(2\pi\sigma^2) - \frac{y_i^2}{2\sigma^2}\right\}.$$

Identify σ^2 as the dispersion parameter and $\theta = \mu$ as the canonical parameter. The cumulant function and variance functions are:

$$b(\theta) = \frac{\theta^2}{2}, \quad \nu(\mu) = \ddot{b}(\theta) = 1.$$

The canonical activation function is the identity:

$$h(\eta) = \dot{b}(\eta) = \eta.$$

Thus $g(\mu) = \mu$ and $\dot{g}(\mu) = 1$.

The characteristic matrices are:

$$\Delta = \mathbf{I}, \quad \mathbf{W} = (\sigma^2)^{-1}\mathbf{I}, \quad \Sigma = \sigma^2\mathbf{I}.$$



4.2 Bernoulli Distribution

Example 4.2 (Canonical Link). The Bernoulli distribution takes the form:

$$f(y_i|\mu_i) = \mu_i^{y_i}(1 - \mu_i)^{1-y_i} = \exp \left\{ y_i \ln \left(\frac{\mu_i}{1 - \mu_i} \right) + \ln(1 - \mu_i) \right\}.$$

Identify:

$$\theta = \ln \left(\frac{\mu}{1 - \mu} \right),$$

as the canonical parameter, which is related to μ via:

$$\mu = \frac{e^\theta}{1 + e^\theta}, \quad 1 - \mu = \frac{1}{1 + e^\theta}.$$

The dispersion parameter is $\phi \equiv 1$.

Expressed in terms of the canonical parameter, the density takes the form:

$$f(y_i|\theta_i) = \exp \{ y_i \theta_i - \ln(1 + e^{\theta_i}) \}.$$

Identify $b(\theta) = \ln(1 + e^\theta)$ as the cumulant function, whose derivatives are:

$$\dot{b}(\theta) = \frac{e^\theta}{1 + e^\theta}, \quad \ddot{b}(\theta) = \frac{e^\theta(1 + e^\theta) - e^\theta(e^\theta)}{(1 + e^\theta)^2} = \frac{e^\theta}{1 + e^\theta} \cdot \frac{1}{1 + e^\theta} = \mu(1 - \mu).$$

The canonical activation function is:

$$h(\eta) = \dot{b}(\eta) = \frac{e^\eta}{1 + e^\eta},$$

with inverse:

$$g(\mu) = \ln \left(\frac{\mu}{1 - \mu} \right) = \ln(\mu) - \ln(1 - \mu).$$

The derivative of the canonical link is:

$$\dot{g}(\mu) = \frac{1}{\mu} + \frac{1}{1 - \mu} = \frac{1}{\mu(1 - \mu)}$$

The characteristic matrices are:

$$\mathbf{\Delta} = \text{diag}\{\mu_i(1 - \mu_i)\}, \quad \mathbf{W} = \text{diag}\left\{ \frac{1}{\mu_i(1 - \mu_i)} \right\}, \quad \mathbf{\Sigma} = \mathbf{W}^{-1}.$$



Example 4.3 (Probit Link). Consider the Bernoulli distribution with the Probit link:

$$g(\mu) = \Phi^{-1}(\mu) = \eta.$$

The first derivative is:

$$\dot{g}(\mu) = \frac{1}{\phi\{\Phi^{-1}(\mu)\}} = \frac{1}{\phi(\eta)},$$

where ϕ denotes the standard normal density. The characteristic matrices become:

$$\Delta = \text{diag} \left\{ \frac{1}{\phi(\eta_i)} \right\}, \quad \mathbf{W} = \text{diag} \left\{ \frac{\phi^2(\eta_i)}{\mu_i(1 - \mu_i)} \right\}, \quad \Sigma = \text{diag} \{ \mu_i(1 - \mu_i) \}.$$

Note that:

$$\mu_i(1 - \mu_i) = \Phi(\eta_i) \{1 - \Phi(\eta_i)\} = \Phi(\eta_i) \Phi(-\eta_i).$$



4.3 Poisson Distribution

Example 4.4 (Canonical Link). The Poisson distribution takes the form:

$$f(y_i | \mu_i) = \frac{e^{-\mu_i} \mu_i^{y_i}}{y_i!} = \exp \{ y_i \ln(\mu_i) - \mu_i - \ln(y_i!) \}.$$

Identify $\theta = \ln(\mu)$ as the canonical parameter, with inverse $\mu = e^\theta$. The dispersion parameter is $\phi \equiv 1$. Expressing the density in terms of the canonical parameter:

$$f(y_i | \theta_i) = \exp \{ y_i \theta_i - e^{\theta_i} - \ln(y_i!) \}.$$

Identify $b(\theta) = e^\theta$ as the cumulant function, with derivatives:

$$\dot{b}(\theta) = \ddot{b}(\theta) = e^\theta.$$

The canonical activation function is $h(\eta) = \dot{b}(\eta) = e^\eta$, with inverse $g(\mu) = \ln(\mu)$. The derivative of the canonical link is $\dot{g}(\mu) = \mu^{-1}$. The characteristic matrices are:

$$\Delta = \text{diag} \left\{ \frac{1}{\mu_i} \right\}, \quad \mathbf{W} = \text{diag} \{ \mu_i \}, \quad \Sigma = \mathbf{W}^{-1}.$$

